

Governance / Constitution Debt

Status Audit

Date: 2026-06-14 **Scope:** Read-only verification of every §6.*-debt entry + CM-* paper-only/owed gate in root CLAUDE.md, cross-checked against the actual code/scripts/hooks at HEAD 357f867a. **Method:** Each row's "actual-status" is verified against a real artifact (file existence, hook line, verify-all wiring), not against the CLAUDE.md prose. Where CLAUDE.md prose disagrees with the artifact, the row is flagged STALE. **Anti-bluff (§11.4.6):** Every claim is cited to a file/line or "absent". No guessed status.

Debt-item status table

debt-id	requires	actual-status	evidence (file:line / absent)
§6.X-debt	Emulator PID inside podman/docker container managed by pkg/runtime; --runner=containerized for gate runs on every OS	PARTIAL	submodules/containers/pkgcontainerized.go exists with kvmDevicePath/kvmAvailable (containerized.go:14-35); b path on macOS still resolves RESOLVED_RUNNER="host-dir (scripts/run-challenge-matrix.sh:198,215). Linux x path works; macOS gate-host Superseded in practice by §6.
§6.Y-debt	Pre-push hook rejects code-touching commit after a distribute that did not bump versionCode	CLOSED	.githubhooks/pre-push:128-15 implements §6.Y bump-first c against last-version-debug. CLAUDE.md still lists it OWE
§6.Z-debt	firebase-distribute Phase-1 Gate 6 (evidence file: matching SHA + ≤24h + BUILD SUCCESSFUL) + pre-push check +	PARTIAL	Pre-push Check 7 enforces ev presence on pointer advance (.githubhooks/pre-push:159-19 scripts/firebase-distribu NO runtime Gate 6 that valid timestamp/BUILD-SUCCESS comment reference (firebas

debt-id	requires	actual-status	evidence (file:line / absent)
	tests/firebase/ hermetic test		distribute.sh:40-41). tests exists but tests version-mono pepper, not the §6.Z evidence (tests/firebase/).
§6.AA-debt	Default --debug-only; reject --release-only without companion debug evidence; split last-version- {debug, release}; tests/firebase/ hermetic test	CLOSED	firebase-distribute.sh:46 MODE="debug" default; :199-2 --release-only when last-v debug < current vc; split po :150, 171. CLAUDE.md still li → STALE.
§6.AB-debt	Detekt/go-vet rule for per-feature anti-bluff completeness (rendering/state- machine/gating)	OPEN	No Detekt rule; config/detekt only detekt.yml (config/detekt listing). The related scripts/ challenge-discrimination. exists+STRICT but covers Cha falsifiability markers, not the completeness checklist.
§6.AC-debt	Detekt rule config/ detekt/lava-non- fatal-required.kts + go-vet lava-api-go/ scripts/check-non- fatal-coverage.sh + pre-push + tests/non- fatal-coverage/	OPEN	config/detekt/lava-non-fa required.kts ABSENT; lava- scripts/check-non-fatal-c ABSENT; tests/non-fatal-c ABSENT. A bash scanner scri non-fatal-coverage.sh exist only, STRICT) but the Go vet custom rule + hermetic tests built.
§6.AD-debt (1) per-submodule pointer propagation	## INHERITED FROM block in all per-scope CLAUDE.md/ AGENTS.md/ CONSTITUTION.md	CLOSED	scripts/check-constitution 6.AD(4) enforces 54 in-scope helix-constitution-gates.
§6.AD-debt (2) CM-COMMIT-DOCS- EXISTS	verify referenced docs in commit messages resolve	CLOSED	scripts/check-commit-docs exists + wired in verify-all (sc verify-all-constitution- rules.sh:190). CLAUDE.md l → STALE.
§6.AD-debt (2) CM-SCRIPT-DOCS- SYNC	script  docs/scripts/ *.md sync gate	CLOSED	scripts/check-script-docs exists + verify-all :187; AND Check 9 (.githubhooks/pre-push CLAUDE.md lists OWED → S

debt-id	requires	actual-status	evidence (file:line / absent)
§6.AD-debt (2) CM-SUBAGENT- DELEGATION-AUDIT	subagent delegation audit trail	CLOSED (scanner)	scripts/check-subagent-de audit.sh exists + verify-all : doc + CLAUDE.md call it paper STALE.
§6.AD-debt (2) CM-COMMIT-DOCS- EXISTS/others equivalence- mapped	CM-ITEM-STATUS/TYPE- TRACKING, CM-FIXED- COLUMN-ALIGNMENT, operator-blocked gates	CLOSED-BY- EQUIVALENCE	§6.AD.3 Path B maps these to evidence/ ledgers (docs/helix- constitution- gates.md:14, 16, 19–22, 36–3 Intentional; no scanner owed
§6.AD-debt (5) build-resource stats tracker	CM-BUILD-RESOURCE- STATS-TRACKER	CLOSED	scripts/build-stats-sampl scripts/build-stats-repor registry exist (docs/helix- constitution-gates.md:18). CLAUDE.md §6.AD-debt-5 st OWED → STALE.
§6.AE-debt	check-challenge- coverage.sh strict- mode flip + per- feature backfill	CLOSED (strict)	scripts/check-challenge- coverage.sh:39 STRICT="\$ {LAVA_CHALLENGE_COVERAGE_ (STRICT default); verify-all in (:163). Gate-doc says "adviso STALE. Runner host-gap is §6 not §6.AE.
§6.AF-debt (2) chaos/stress §11.4.85 (LVA-7)	runnable chaos/stress beyond phase-1 lava- api-go	PARTIAL	scripts/run-chaos-stress. lines, scaffold); docs/chaos-s DESIGN.md + EVIDENCE-phase NOT wired into verify-all (no in scripts/verify-all-cons rules.sh). Phase-2+ owed.
§6.AF-debt (LVA-6) codegraph own- org submodule inclusion	§11.4.79 — index own- org submodule SOURCE	CLOSED	.codegraph/config.json:3 _ records LVA-6 CLOSED 2026- (1,842 submodule files index submodule probe resolves); b submodules/** exclude remo CLAUDE.md §6.AF lists OWE
§6.AF-debt (4) per-clause CM- COVENANT-114-* -PROPAGATION gates	literal-anchor scan gates for §11.4.79–106	OPEN	Only CM-COVENANT-114-16-PP row present (docs/helix-con gates.md:14); no per-clause scanner for the new 79–106 a
§6.AF-debt (5) §11.4.80	auto git submodule init before index	OPEN	Precondition documented in .codegraph/config.json:3 b automation script that runs g

debt-id	requires	actual-status	evidence (file:line / absent)
codegraph sync automation			submodule init before codegraph index.
§6.AG (no separate -debt)	Containers-driven emulators, no live ADB device	PARTIAL	Policy is in CLAUDE.md §6.AG enforced via run-challenge- but the no-host-direct requirement superseded/tightened by §6.AH blocked by §6.AH-debt.
§6.AH-debt	Container/VM emulator path that BOOTS on macOS (no-KVM TCG mode + macOS /proc-reap fix in Containers submodule); no host-direct fallback	OPEN	containerized.go hard-require kvm for acceleration (containerized.go:19-21;34 no-KVM software-emulation challenge-matrix.sh:215 still macOS→host-direct. Incidents .lava-ci-evidence/sixth-lava incidents/2026-06-03-emulation offline.json (present). §6.Z canary + client gate honestly on this macOS host.
§6.AI-debt (1) §11.4.128 device recorder	background Genymotion/emulator recorder w/ deterministic layout	PARTIAL/ CLOSED?	scripts/record-device-session exists; tests/device-record Whether it implements the fully deterministic YYYY-MM-DD/<session hash>/... layout + gitignore, exclude was not fully verified PARTIAL pending deeper checks
§6.AI-debt (2) §11.4.140 LAYER 2 action-prefix hook	UserPromptSubmit hook + generated slash commands	CLOSED (hook)	scripts/hooks/action-prefix expand.sh exists + wired in . settings.json:14-22 (UserPromptSubmit); constitution actions/registry.yaml present Generated per-agent slash commands not verified. CLAUDE.md lists STALE for the hook portion.
§6.AI-debt (3) §11.4.141 thin-index restructure + token harness	de-dup 90K-token CLAUDE.md to index lines	OPEN	No evidence of a thin-index root CLAUDE.md remains a full document.
§6.AI-debt (4) per-clause CM-COVENANT-114-{128..141}-	literal-anchor gates for §11.4.128-141	OPEN	Not present in docs/helix-constitution-gates.md gate

debt-id	requires	actual-status	evidence (file:line / absent)
PROPAGATION gates			
CM-COVERAGE- LEDGER	§11.4.25 — strict-flip after baseline (Phase 7-debt)	CLOSED (strict)	scripts/check-coverage-le STRICT=\$ {LAVA_COVERAGE_LEDGER_STR verify-all :156 invokes --str doc + CLAUDE.md mark it "ac STRICT flip OWED" → STALE
CM-LVA-TICKETS- SYNC → CM- WORKABLE-ITEMS- SYNC	canonical workable- items binary; strict- flip	PARTIAL	scripts/check-workable-it exists + verify-all :202. Gate- names the retired CM-LVA-TI scripts/check-lva-tickets helix-constitution-gates. STALE; CLAUDE.md §6.AF no migration to canonical done. update/reopen/block + expor OWED (non-blocking).
CM- COVENANT-114-16- PROPAGATION + item-tracking paper-only gates	§11.4.15/16/19/21 trackers	CLOSED-BY- EQUIVALENCE	Intentional Path-B mapping helix-constitution- gates.md:14,19-24). Not true

STALE-claim findings (CLAUDE.md/gate-doc says OPEN but artifact is CLOSED)

These are documentation drift, not real work. A master plan should NOT
schedule work for them — only doc corrections:

1. **§6.Y-debt** — pre-push Check 6 is live (.githubhooks/pre-push:128).
CLAUDE.md lists OWED.
2. **§6.AA-debt** — default flipped to --debug-only; release-only companion
check live (firebase-distribute.sh:46,199). CLAUDE.md lists
PARTIAL.
3. **§6.AD-debt items 2** — check-commit-docs-exists.sh, check-script-
docs-sync.sh, check-subagent-delegation-audit.sh all exist + verify-
all-wired + pre-push Check 9. Gate-doc marks several ⚠ paper-only.
4. **§6.AD-debt item 5** — build-stats tracker shipped (build-stats-
sample.sh/build-stats-report.sh). CLAUDE.md still lists OWED.
5. **§6.AE-debt** — check-challenge-coverage.sh is STRICT-default (:39).
Gate-doc says advisory.

6. **CM-COVERAGE-LEDGER** — STRICT-default (check-coverage-ledger.sh:41, verify-all --strict). CLAUDE.md + gate-doc say advisory / Phase-7-debt OWED.
7. **LVA-6 (codegraph submodules)** — CLOSED per .codegraph/config.json:3. CLAUDE.md §6.AF lists OWED.
8. **§6.AI LAYER 2 action hook** — wired in .claude/settings.json:14. CLAUDE.md lists OWED.
9. **docs/helix-constitution-gates.md itself is stale** — "Last updated 2026-05-16"; predates §6.AF/AG/AH/AI (May 31 – Jun 9). Several rows describe pre-strict-flip state.

Truly-open debts (real work to schedule)

debt-id	one-line	effort	blocking
§6.AH-debt	macOS container/VM (TCG no-KVM) emulator path that actually boots; no host-direct fallback	L	YES
§6.X-debt	container-bound emulator gate path on macOS (folds into §6.AH)	L	YES
§6.AC-debt	Detekt lava-non-fatal-required.kts + Go-vet check-non-fatal-coverage.sh + tests/non-fatal-coverage/	M	no
§6.AB-debt	Detekt/go-vet per-feature anti-bluff completeness rule	M	no
§6.Z-debt (residual)	runtime Gate 6 inside firebase-distribute.sh (SHA/24h/BUILD-SUCCESSFUL) + hermetic test	M	partial
§6.AF-debt chaos (LVA-7)	chaos/stress §11.4.85 beyond phase-1; wire into verify-all	M	no
§6.AI-debt §11.4.141	thin-index restructure of CLAUDE.md + token-reduction harness	L	no
§6.AF/AI per-clause CM-COVENANT-**-PROPAGATION gates	literal-anchor scanners for §11.4.79–141	S	no
		S	no

debt-id	one-line	effort	blocking
§6.AF-debt §11.4.80	auto git submodule init before codegraph index		
§6.AI-debt §11.4.128	verify/complete deterministic device-recorder layout	S	no

Prioritized recommendations

1. **§6.AH-debt + §6.X-debt (TOP — both BLOCKING)**. Single coordinated effort in the Containers submodule: add a no-KVM TCG software-emulation runner + the macOS /proc-reap fix so the container/VM emulator path boots on darwin/arm64. Until this lands, EVERY §6.AE matrix gate, §6.Z release canary, and tag-time attestation on the operator's macOS host is honestly BLOCKED — this is the single highest-leverage debt because it unblocks the entire release pipeline.
2. **§6.Z-debt residual (firebase Gate 6)**. Add the runtime evidence gate inside `firebase-distribute.sh` (the pre-push Check 7 only fires on the commit that advances the pointer, not at distribute time). Cheaper than #1 and closes a real distribute-faulty-version vector.
3. **§6.AC-debt (non-fatal telemetry lint)**. Build the Detekt custom rule + Go-vet check + `tests/non-fatal-coverage/`. Currently only a Kotlin bash scanner exists; the Go side has zero enforcement. Pairs naturally with §6.AB-debt (same Detekt-rule-authoring effort).

Doc-hygiene action (not scheduled work, but should precede the plan): correct the 9 STALE claims above in `CLAUDE.md` + regenerate `docs/helix-constitution-gates.md` (stale since 2026-05-16) so the master plan is built on accurate status. Several debts are already done.

Open vs closed count: Of ~27 distinct debt/gate items examined: **10 truly-open** (2 BLOCKING, 1 partial-blocking, 7 non-blocking), **3 PARTIAL**, ****~14 CLOSED** or closed-by-equivalence** (9 of which are STALE-marked-open in `CLAUDE.md/gate-doc`).